

Airline Satisfaction Analysis

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Data Dictionary

1	Field	Description
2	ID	Unique passenger identifier
3	Gender	Gender of the passenger (Female/Male)
4	Age	Age of the passenger
5	Customer Type	Type of airline customer (First-time/Returning)
6	Type of Travel	Purpose of the flight (Business/Personal)
7	Class	Travel class in the airplane for the passenger seat
8	Flight Distance	Flight distance in miles
9	Departure Delay	Flight departure delay in minutes
10	Arrival Delay	Flight arrival delay in minutes
11	Departure and Arrival Time Convenience	Satisfaction level with the convenience of the flight departure and arrival times from 1 (lowest) to 5 (highest) - 0 means "not applicable"
12	Ease of Online Booking	Satisfaction level with the online booking experience from 1 (lowest) to 5 (highest) - 0 means "not applicable"
13	Check-in Service	Satisfaction level with the check-in service from 1 (lowest) to 5 (highest) - 0 means "not applicable"
14	Online Boarding	Satisfaction level with the online boarding experience from 1 (lowest) to 5 (highest) - 0 means "not applicable"
15	Gate Location	Satisfaction level with the gate location in the airport from 1 (lowest) to 5 (highest) - 0 means "not applicable"
16	On-board Service	Satisfaction level with the on-boarding service in the airport from 1 (lowest) to 5 (highest) - 0 means "not applicable"
17	Seat Comfort	Satisfaction level with the comfort of the airplane seat from 1 (lowest) to 5 (highest) - 0 means "not applicable"
18	Leg Room Service	Satisfaction level with the leg room of the airplane seat from 1 (lowest) to 5 (highest) - 0 means "not applicable"
19	Cleanliness	Satisfaction level with the cleanliness of the airplane from 1 (lowest) to 5 (highest) - 0 means "not applicable"
20	Food and Drink	Satisfaction level with the food and drinks on the airplane from 1 (lowest) to 5 (highest) - 0 means "not applicable"
21	In-flight Service	Satisfaction level with the in-flight service from 1 (lowest) to 5 (highest) - 0 means "not applicable"
22	In-flight Wifi Service	Satisfaction level with the in-flight Wifi service from 1 (lowest) to 5 (highest) - 0 means "not applicable"
23	In-flight Entertainment	Satisfaction level with the in-flight entertainment from 1 (lowest) to 5 (highest) - 0 means "not applicable"
24	Baggage Handling	Satisfaction level with the baggage handling from the airline from 1 (lowest) to 5 (highest) - 0 means "not applicable"
25	Satisfaction	Overall satisfaction level with the airline (Satisfied/Neutral or unsatisfied)

Data Information

Field	Type
Gender	Categorical
Customer Type	Categorical
Type of Travel	Categorical
Class	Categorical
Satisfaction	Categorical
Age	Numerical
ID	Numerical
Flight Distance	Numerical
Departure Delay	Numerical
Arrival Delay	Numerical
Departure and Arrival Time Convenience	Numerical
Ease of Online Booking	Numerical
Check-in Service	Numerical
Online Boarding	Numerical
Gate Location	Numerical
On-board Service	Numerical
Seat Comfort	Numerical
Leg Room Service	Numerical
Cleanliness	Numerical
Food and Drink	Numerical
In-flight Service	Numerical
In-flight Wifi Service	Numerical
In-flight Entertainment	Numerical
Baggage Handling	Numerical

	Feature	Mode	Mean	Median	Dispersion	Min.	Max.	Missing
1	ID	1	64940.5	64940.5	0.577346	1	129880	0
2	Age	39	39.428	40	0.383467	7	85	0
3	Flight Distance	337	1190.32	844	0.837969	31	4983	0
4	Departure D...	0	14.7137	0	2.58745	0	1592	0
5	Arrival Delay	0	15.0911	0	2.54888	0	1584	393
6	Departure an...	4	3.0576	3	0.499325	0	5	0
7	Ease of Onlin...	3	2.75688	3	0.50845	0	5	0
8	Check-in Ser...	4	3.30627	3	0.382964	0	5	0
9	Online Board...	4	3.25263	3	0.415268	0	5	0
10	Gate Location	3	2.97692	3	0.429475	0	5	0
11	On-board Se...	4	3.38302	4	0.380457	0	5	0
12	Seat Comfort	4	3.44136	4	0.383361	0	5	0
13	Leg Room S...	4	3.35088	4	0.392807	0	5	0
14	Cleanliness	4	3.28633	3	0.39974	0	5	0
15	Food and Dri...	4	3.20477	3	0.414983	0	5	0
16	In-flight Serv...	4	3.64219	4	0.323065	0	5	0
17	In-flight Wifi ...	2	2.7287	3	0.487169	0	5	0
18	In-flight Ente...	4	3.35808	4	0.397264	0	5	0
19	Baggage Ha...	4	3.63211	4	0.324885	1	5	0

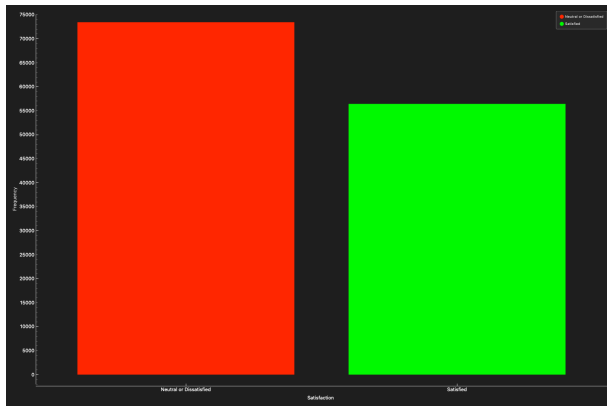
Variables:

- Categorical: 5
- Numerical: 19

Observations for numerical variables

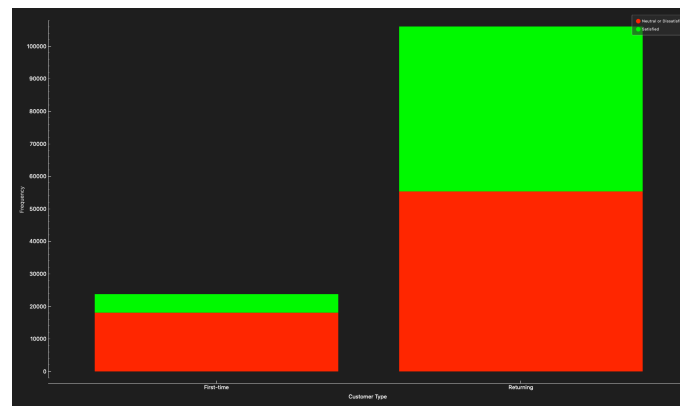
- Flight distances vary widely, with a **median of 844 miles** and a **maximum reaching 4,983 miles**, highlighting the presence of long-haul routes. The most frequent distance traveled is **337 miles**, indicating that short-to-medium-haul flights are common.
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- For **departure delays**, the **most common scenario is no delay** (mode = 0, median = 0), but in some cases, delays stretch up to **1,592 minutes**. The **average delay is 14.71 minutes**, suggesting occasional disruptions. **Arrival delays mirror departure delays**, with most flights arriving punctually but a few experiencing significant lateness.
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- Passengers generally find **departure and arrival times convenient**, as the **mode is 4**, meaning most travelers are satisfied with scheduling. **Online booking ease receives mixed feedback**, with an **average rating of around 2.75**, implying some users struggle with the process.
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- The **check-in experience is well-rated**, with the most common rating being **4**. Similarly, **online boarding is viewed positively**, with a majority giving it a **4-star rating**. However, the **overall airline rating is slightly below 3**, suggesting room for improvement.
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- For **onboard services**, passengers commonly rate them as **good** (mode = 4), and **comfort levels are also rated positively**, aligning with seat comfort scores. **Cleanliness is rated fairly well**, though it ranks slightly below seat comfort and onboard service.
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- However, **food quality is considered unsatisfactory**, standing out as a weak point. **In-flight service receives one of the highest ratings**, while **in-flight Wi-Fi is among the lowest-rated aspects**. **In-flight entertainment is rated similarly to onboard service**, indicating general approval.

Which percentage of airline passengers are satisfied? Does it vary by customer type? What about the kind of travel?



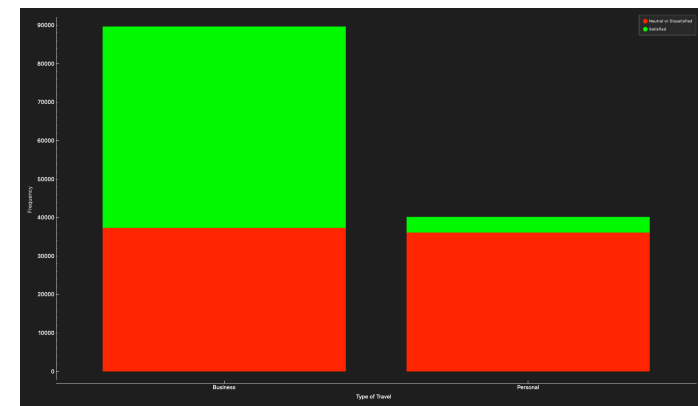
Observations:

- 43.45% of total customers are satisfied
- 56.56% of total customers are neutral or dissatisfied



Observations:

- 23.97% of first-time customers are satisfied (4.39% of overall customers)
- 47.81% of returning customers are satisfied (39.06% of overall customers)



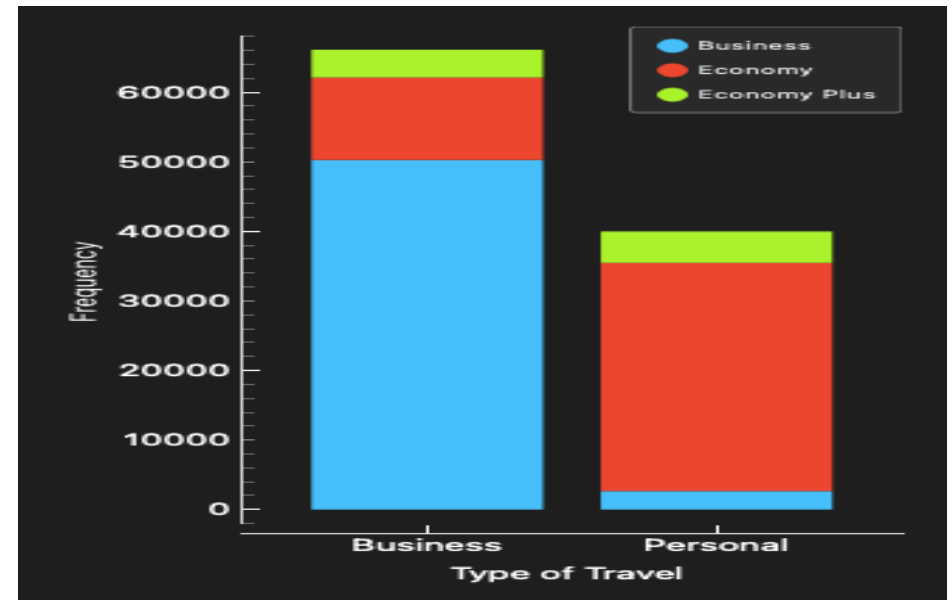
Observations:

- 10.13% of personal travel customers are satisfied (3.14% of overall customers)
- 58.37% of business travel customers are satisfied (40.31% of overall customers)

What is the customer profile for a repeating airline passenger?

The average repeating airline passenger profile is:

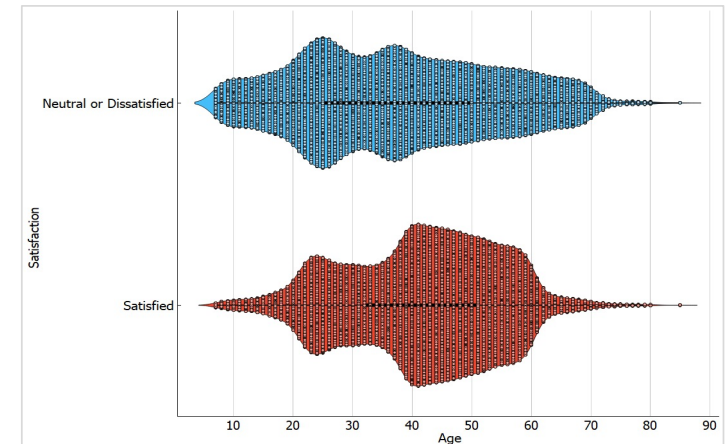
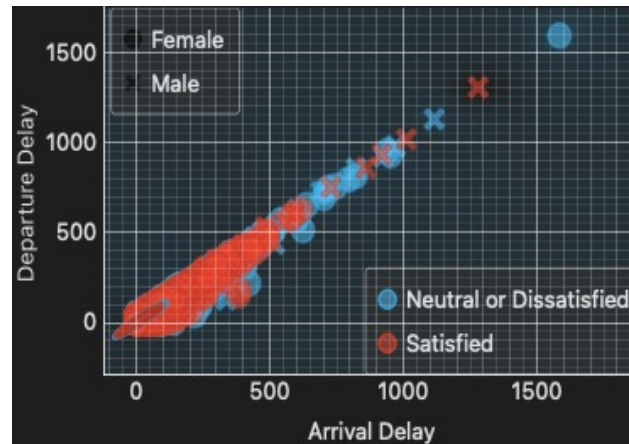
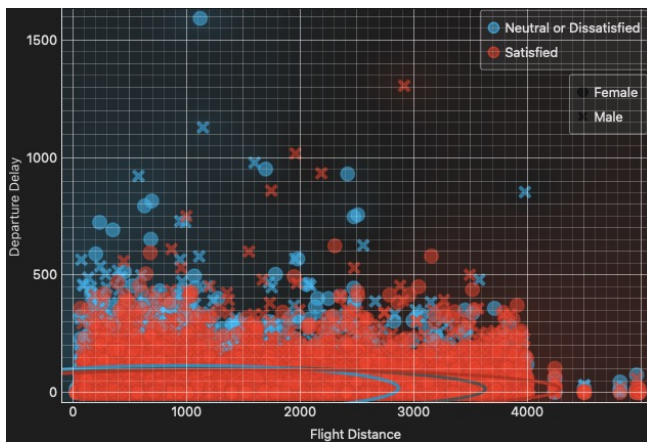
- *43 years old*
- *Female*
- *Travels for business*
- *Purchases a business class seat*
- *Experiences flight delays infrequently*
- *Finds seat comfort and leg room to be satisfactory*
- *Avg Flight Distance is about 337 miles*



Observations:

- *Repeating customers traveling for business are significantly more likely to purchase business class tickets*
- *Repeating customers traveling for personal primarily purchase economy tickets*

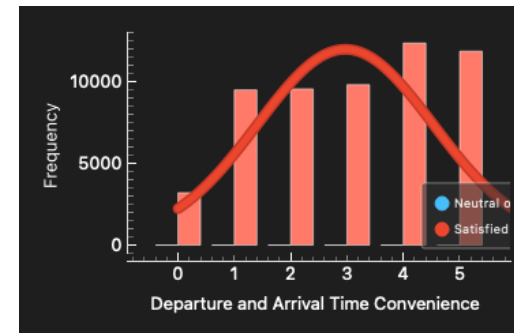
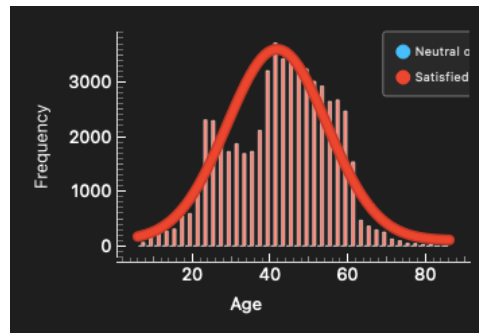
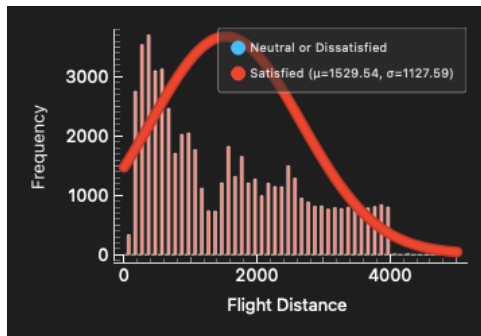
Does flight distance affect customer preferences or flight patterns?



Observations:

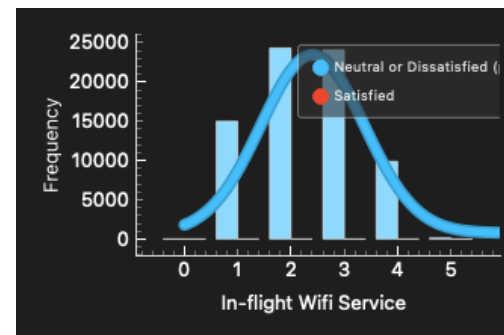
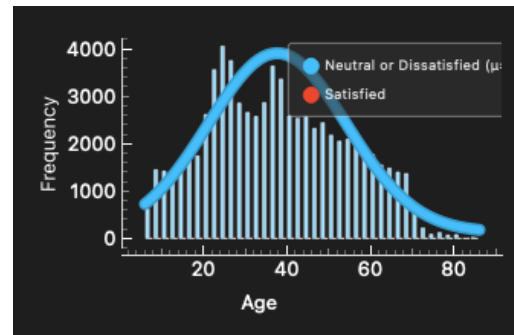
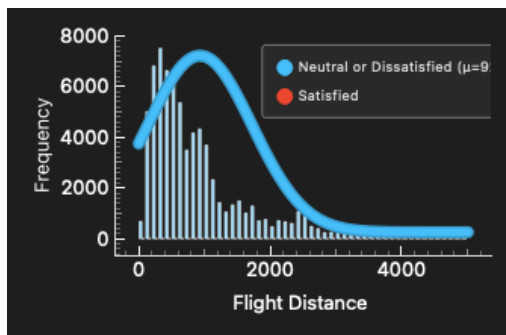
- Customers are more likely to be satisfied when flight distance is greater than 2,000 miles
- Customers are more likely to remain satisfied if delays are less than 600 minutes
- Younger customers are more likely to be neutral or dissatisfied

Which factors contribute to customer satisfaction the most? What about dissatisfaction?



The following factors contribute to customer satisfaction:

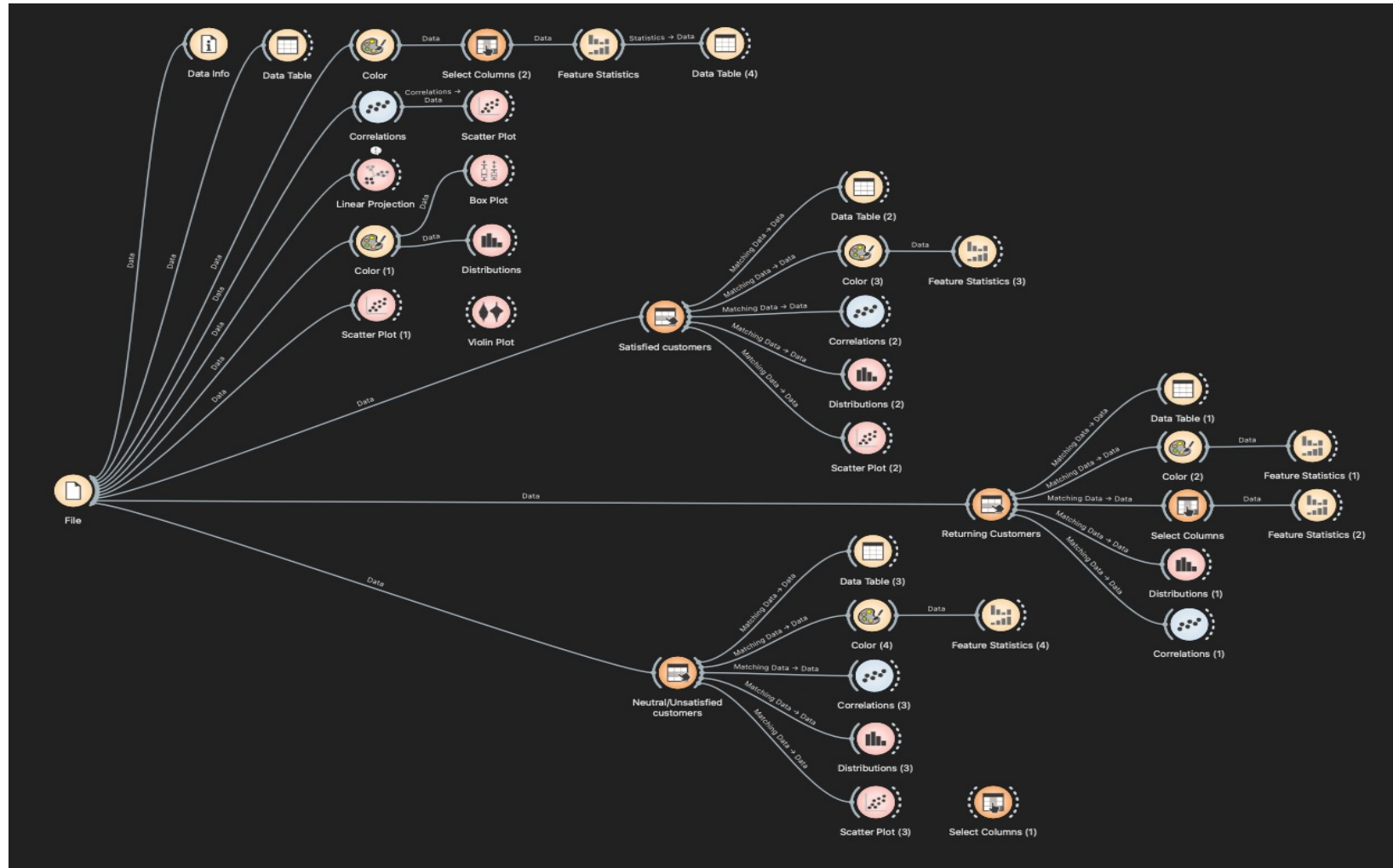
- Flight distances
- Older Passengers
- Departure and arrival time convenience



The following factors contribute to customer dissatisfaction:

- Flight distance
- Age
- Ease of online booking
- In Flight Services

Orange Workflow



Final Observations

1. Convert First-Time Young Travelers into Loyal Customers

Younger passengers are more likely to be first-time flyers. Airlines can **increase retention** by offering targeted loyalty programs, digital perks, and seamless experiences tailored to tech-savvy travelers.

2. Enhance Comfort for Economy Class

Economy passengers often feel cramped and uncomfortable, leading to dissatisfaction. Improving **seat padding, legroom, and in-flight service** could significantly enhance their experience, making them more likely to return.

3. Minimize Flight Delays

Delays remain one of the biggest contributors to customer dissatisfaction. Keeping delays **under 600 minutes** and optimizing **schedules, turnarounds, and real-time communication** can greatly reduce frustration.

4. Business Class & Business Travelers Show Higher Satisfaction

Business-class passengers are significantly more likely to be satisfied and return. Similarly, business travelers tend to have a better overall experience compared to leisure travelers. Investing in **premium perks and efficient services** can further boost satisfaction.

5. Speed Up Check-in & Baggage Handling

Long check-in lines and lost baggage create unnecessary stress. **Streamlining check-in processes and improving baggage tracking** will enhance the passenger experience and reduce complaints.

6. Improve Short-Haul Flight Services

Passengers on short-haul flights often feel underserved. Simple improvements like **better snacks, smoother boarding, and onboard entertainment** can make these flights more enjoyable.

7. Cater to Younger Travelers' Expectations

Younger passengers expect **fast Wi-Fi, engaging in-flight entertainment, and seamless digital interactions**. Investing in **stronger internet, app-based services, and tech-driven enhancements** can help airlines build loyalty among this demographic.